

Disproof of Conjecture 00000007739: the fourth moment of a free semicircular word of length one is 2, not 9/4

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Abstract

Conjecture 00000007739 asserts that for free words W of length d in a free semicircular family, the ratio $M_p(d) = \|W\|_p / \|W\|_2$ satisfies $M_4(d)^2 = 2 - d^{-1}(1 - 2^{-d})$. This is false already for $d = 1$. A word of length one is a single standard semicircular element s , with $\tau(s^2) = 1$ and $\tau(s^4) = 2$, so $M_4(1)^4 = 2$ and $M_4(1)^2 = \sqrt{2}$, whereas the formula gives $3/2$ and $(3/2)^2 = 9/4 \neq 2$. The same value $M_4(1)^4 = 2$ holds for *every* nonzero element of the first semicircular chaos (real or complex coefficients), so reading $M_p(d)$ as an optimal (or extremal) constant changes nothing. More generally, for $W = s_1 \cdots s_d$ we have $M_4(d)^4 = d + 1$, and the formula fails for every $d \geq 1$ and under each of the normalisations M_4 , M_4^2 , M_4^4 . The decisive objects (Voiculescu's semicircular elements $\ell(e_i) + \ell(e_i)^*$ on the full Fock space, the vacuum state, adjoints and the norms) are constructed from scratch in Lean 4 (core library only), and the negation of the clause is proved there. An independent Python script recomputes all moments exactly by two methods.

1 The claim

The conjecture (English version): “*The free Khintchine inequalities compare L^p and L^2 norms of noncommutative martingale polynomials: for free words W of length d generated by free semicircular families, $M_p(d) = \|W\|_p / \|W\|_2$. Conjecture: $M_4(d)^2 = 2 - d^{-1}(1 - (1/2)^d)$, the exact free refinement of the fourth-moment theorem; and the limit ratio spectrum of $M_\infty(d)/M_p(d)$ as d grows characterizes the operator-norm concentration of products of free group generators.*” The Chinese version calls $M_p(d)$ “the optimal constant of the ratio”.

The statement is a conjunction of

(C1) $M_4(d)^2 = f(d)$ for every $d \geq 1$, where

$$f(d) := 2 - \frac{1 - 2^{-d}}{d} = \frac{2d \cdot 2^d - 2^d + 1}{d \cdot 2^d},$$

so that $f(1) = \frac{3}{2}$, $f(2) = \frac{13}{8}$, $f(3) = \frac{41}{24}$, $f(4) = \frac{113}{64}$ and $f(5) = \frac{289}{160}$;

(C2) an informal statement on $M_\infty(d)/M_p(d)$ and free group generators.

We refute (C1); hence the conjunction is false whatever (C2) means. Note $\frac{3}{2} \leq f(d) < 2$ for all $d \geq 1$ (since $0 < (1 - 2^{-d})/d \leq \frac{1}{2}$).

2 Set-up

Let $(\mathcal{A}, \tau)$ be a tracial W^* -probability space. For $x \in \mathcal{A}$ and $p < \infty$, $\|x\|_p = \tau(|x|^p)^{1/p}$ with $|x| = (x^*x)^{1/2}$. In particular

$$\|x\|_2^2 = \tau(x^*x), \quad \|x\|_4^4 = \tau((x^*x)^2).$$

The ratio $\|x\|_4/\|x\|_2$ is invariant under $x \mapsto cx$ ($c \neq 0$).

Full Fock space (Voiculescu). Let H be a Hilbert space with orthonormal basis $(e_i)_{i \geq 0}$ and $\mathcal{F}(H) = \mathbb{C}\Omega \oplus \bigoplus_{n \geq 1} H^{\otimes n}$, with orthonormal basis $e_{i_1} \otimes \cdots \otimes e_{i_n}$ (the empty word is the vacuum Ω). The creation operator $\ell(e_i)\xi = e_i \otimes \xi$ has adjoint $\ell(e_i)^*$, given by $\ell(e_i)^*\Omega = 0$ and $\ell(e_i)^*(e_j \otimes \xi) = \delta_{ij}\xi$. Put

$$X_i = \ell(e_i) + \ell(e_i)^*, \quad \tau(T) = \langle \Omega, T\Omega \rangle.$$

By Voiculescu's theorem, $X_0, X_1, \dots$ are free standard semicircular elements with respect to τ , and τ is a faithful trace on the von Neumann algebra they generate. Since any two free semicircular families of the same size have the same joint $*$ -distribution, every quantity below is the same in every realisation. For a self-adjoint y in this algebra, $\tau(y^2) = \|y\Omega\|^2$. Hence, for every x ,

$$\|x\|_2^2 = \|x\Omega\|^2, \quad \|x\|_4^4 = \|x^*x\Omega\|^2. \quad (1)$$

Moment formula. For the free semicircular family, $\tau(X_{i_1} \cdots X_{i_m})$ is the number of non-crossing pair partitions $\pi \in \text{NC}_2(m)$ such that i is constant on each block of π (Speicher's free Wick formula; it also follows directly by expanding the product of ℓ 's and ℓ^* 's applied to Ω).

3 The decisive case $d = 1$

Theorem 1. *Let s be a standard semicircular element. Then $\|s\|_2^2 = 1$ and $\|s\|_4^4 = 2$, so $M_4(1)^4 = 2$. Consequently $M_4(1)^2 = \sqrt{2} \neq \frac{3}{2} = f(1)$, and clause (C1) is false.*

Proof. Realise $s = X_0$. Then $X_0\Omega = e_0$ and $X_0^2\Omega = X_0e_0 = e_0 \otimes e_0 + \Omega$. By (1), $\|s\|_2^2 = \|e_0\|^2 = 1$ and $\|s\|_4^4 = \|e_0 \otimes e_0 + \Omega\|^2 = 1 + 1 = 2$. (Equivalently, there are two non-crossing pairings of four points.) If $M_4(1)^2 = f(1) = 3/2$, then squaring gives $2 = M_4(1)^4 = 9/4$, which is false. $\square$

A word of length one in a free semicircular family is a single s_i , so for $d = 1$ there is no ambiguity about which word is meant.

4 Optimal-constant reading: the whole first chaos

The Chinese text suggests that $M_p(d)$ is the *optimal constant*, i.e. the supremum of $\|W\|_p/\|W\|_2$ over a class of degree- d elements. For $d = 1$ the natural class is the first chaos $\{\sum_i c_i X_i\}$ (finitely many nonzero c_i , or $c \in \ell^2$ by continuity), with real or complex coefficients.

Proposition 2. *For every nonzero $S = \sum_i c_i X_i$ with $c_i \in \mathbb{C}$: $\|S\|_4^4 = 2\|S\|_2^4$. Hence the ratio $\|S\|_4/\|S\|_2 = 2^{1/4}$ is constant on the first chaos. Its supremum, infimum and every value it takes give $M_4(1)^4 = 2$.*

Proof. Let $\beta = \sum_i |c_i|^2$. Then $S\Omega = \sum_j c_j e_j =: \xi$, so $\|S\|_2^2 = \beta$. Next $S^* = \sum_i \bar{c}_i X_i$, and $\ell(e_i)^* \xi = c_i \Omega$, so

$$S^* S \Omega = \sum_i \bar{c}_i (e_i \otimes \xi + c_i \Omega) = \sum_{i,j} \bar{c}_i c_j e_i \otimes e_j + \beta \Omega.$$

The two summands are orthogonal, so $\|S^* S \Omega\|^2 = \sum_{i,j} |c_i|^2 |c_j|^2 + \beta^2 = 2\beta^2$, and (1) finishes the proof. $\square$

For real coefficients this is the stability of the semicircle law: $\sum c_i X_i$ is semicircular of variance $\sum c_i^2$. For complex coefficients the circular element $(X_0 + iX_1)/\sqrt{2}$ and the elliptic elements $aX_0 + ibX_1$ are covered. Thus under the optimal-constant reading, real or complex, $M_4(1)^2 = \sqrt{2} \neq 3/2$ again.

5 General d

Proposition 3. *Let $W_d = s_1 s_2 \cdots s_d$ with $s_1, \dots, s_d$ free standard semicirculars. Then $\|W_d\|_2^2 = 1$ and $\|W_d\|_4^4 = d + 1$.*

Proof. $\tau(W_d^* W_d) = \tau(s_d \cdots s_1 s_1 \cdots s_d)$. Every colour occurs twice, and the only admissible pairing pairs the two occurrences of each colour. This pairing is non-crossing (the pairs are nested), so the moment is 1.

For $\tau((W_d^* W_d)^2)$, write the word as $A B C D$ with $A = C = (d, \dots, 1)$ and $B = D = (1, \dots, d)$. Colour k occurs at positions $a_k = d + 1 - k$, $b_k = d + k$, $c_k = 3d + 1 - k$ and $d_k = 3d + k$. A monochromatic pairing pairs, for each k , these four points as $(a_k b_k)(c_k d_k)$ (“inner”), $(a_k d_k)(b_k c_k)$ (“outer”), or $(a_k c_k)(b_k d_k)$. The last is crossing. For $k < l$:

- if k is outer and l is inner, then $a_l < a_k < b_l < d_k$, so $(a_l b_l)$ crosses $(a_k d_k)$;
- in the three other cases the pairs are nested or disjoint. If both are inner, $(a_k b_k)$ lies inside $(a_l b_l)$ and $(c_k d_k)$ inside $(c_l d_l)$. If k is inner and l is outer, $(a_k b_k)$ and $(c_k d_k)$ lie inside $(a_l d_l)$ and outside $(b_l c_l)$. If both are outer, $(a_k d_k) \supset (b_l c_l)$ and $(b_k c_k) \supset (b_l c_l)$, all inside $(a_l d_l)$.

So the non-crossing monochromatic pairings correspond to the sets of outer colours that are up-sets $\{m, \dots, d\}$ of $\{1, \dots, d\}$, $m \in \{1, \dots, d + 1\}$. There are $d + 1$ of them. (This is the Fuss–Catalan number $\frac{1}{2d+1} \binom{2d+2}{2}$, the second moment of the free multiplicative convolution of d free Poisson laws.) $\square$

Corollary 4. *For every $d \geq 1$ the word W_d violates (C1), and also the variants $M_4(d) = f(d)$ and $M_4(d)^4 = f(d)$.*

Proof. $M_4(d)^4 = d + 1$ is an integer. Each variant, raised to the appropriate power, would give $d + 1 = f(d)^k$ with $k \in \{1, 2, 4\}$. But the numerator $2d \cdot 2^d - 2^d + 1$ of $f(d)$ is odd, so the 2-adic valuation is $v_2(f(d)) = -d - v_2(d) < 0$, and $f(d)^k$ is never an integer. $\square$

Directly: $M_4(d)^2 = \sqrt{d+1} \geq \sqrt{2}$, while $f(d) < 2$. Equality would require $d + 1 = f(d)^2 \in [9/4, 4)$, i.e. $d = 2$, but $f(2)^2 = 169/64 \neq 3$.

6 Readings covered

- **Normalisation.** At $d = 1$: $M_4(1) = 2^{1/4}$, $M_4(1)^2 = \sqrt{2}$ and $M_4(1)^4 = 2$. The formula value $3/2$ matches none of them: $(3/2)^4 = 81/16 \neq 2$, $(3/2)^2 = 9/4 \neq 2$ and $3/2 \neq 2$. For general d see Corollary 4.
- **Which word.** For $d = 1$ every word is a single semicircular. For $d \geq 2$ we use the word with distinct letters, $s_1 \cdots s_d$. Words with repeated letters give other values (e.g. $\|s^2\|_4^4 / \|s^2\|_2^4 = 14/4$), so $M_p(d)$ is then only meaningful as an extremal constant.
- **Optimal constant (sup), or inf, over the d -th chaos; real or complex coefficients.** For $d = 1$ the ratio is constantly 2 (Proposition 2). For $d \geq 2$: the d -th Wigner chaos is $\{x : x\Omega \in H^{\otimes d}\}$. For such x and $\xi \in H^{\otimes d}$, the free Wick formula $x = \sum_k \ell(\cdot)^k \ell(\cdot)^{*d-k}$ (Biane–Speicher) shows that the $H^{\otimes 2d}$ -component of $x^* \xi$ is $(x^* \Omega) \otimes \xi$ and its Ω -component is $\langle x\Omega, \xi \rangle \Omega$. Taking $\xi = x\Omega$ and using traciality gives $\|x\|_4^4 = \|x^* x \Omega\|^2 \geq 2 \|x\|_2^4$ (for self-adjoint x this is the bound of Kemp–Nourdin–Peccati–Speicher). So the ratio $\|x\|_4^4 / \|x\|_2^4$ is ≥ 2 , its supremum is $\geq d + 1$ (Proposition 3), and its infimum is exactly 2. Indeed, for n free copies $w_1, \dots, w_n$ of $s_1 \cdots s_d$, the element $\sum_j w_j$ has ratio $2 + (d - 1)/n$ (`verify.py` checks this). Since $9/4 \leq f(d)^2 < 4$ and $f(2)^2 = 169/64 < 3$, neither the supremum nor the infimum equals $f(d)^2$ for any $d \geq 1$. Since $f(d) < 2$, no element has ratio $f(d)$ (variant $M_4^4 = f$). In the variant $M_4 = f$, the infimum 2 differs from $f(d)^4 \geq 81/16$. We did not determine the exact supremum for $2 \leq d \leq 14$ in that variant. This is not needed, because $d = 1$ already refutes the clause in every variant.
- **Haar unitaries / free group generators** (suggested by (C2)). A word in free Haar unitaries $u_i = \lambda(g_i)$ is unitary, so $|W| = 1$ and $M_p(d) = 1$ for all p ; $1 \neq f(d)$, $f(d)^2$, $f(d)^4$. For linear combinations $x = \sum_s a_s \lambda(s)$ over generators and their inverses, the identity $s^{-1} t u^{-1} v = e$ holds iff $(s = t, u = v)$ or $(t = u, s = v)$. Hence

$$\frac{\|x\|_4^4}{\|x\|_2^4} = 2 - \frac{\sum_s |a_s|^4}{(\sum_s |a_s|^2)^2} \in [1, 2).$$

So $M_4^2 < \sqrt{2} < 3/2$ and $M_4 < 3/2$. In the variant $M_4^4 = f(1) = 3/2$, this value *is* attained, by $u + u^*$ (the arcsine law: $\tau((u + u^*)^4) / \tau((u + u^*)^2)^2 = 6/4$). But it is not the optimal constant, which is $\sup = 2$ (or $\inf = 1$). Moreover the conjecture explicitly concerns *semicircular* families, where the value is 2.

7 Machine verification

Lean 4 (`lean4/Main.lean`, **core only**, **v4.19.0**). Everything is defined from scratch:

- basis words `Word := List Nat` of the full Fock space, and finitely supported vectors `Vec := List (Word × Int)` (formal finite combinations; `coeff`, `inner`). All vectors that occur have integer coordinates.
- `create i = ℓ(e_i)`, `annih i = ℓ(e_i)*`, `X i = ℓ(e_i) + ℓ(e_i)*`, the vacuum Ω , and the state `tau ws = ⟨Ω, Xi1 ⋯ Xik Ω⟩`.
- adjointness: `inner_create`, `inner_annih` (so $\ell(e_i)^*$ is the adjoint of $\ell(e_i)$ on finitely supported vectors), `X_selfadjoint`, and `applyW_adjoint`: the adjoint of $X_{i_1} \cdots X_{i_k}$ is $X_{i_k} \cdots X_{i_1}$.

- `norm2sq` = $\tau(W^*W)$ and `norm4pow4` = $\tau((W^*W)^2)$, with `norm2sq_eq`, `norm4pow4_eq` proving (1).

Results:

- `semicircle_moments`: $\tau(X_0^k)$ for $k \leq 10$ are 1, 0, 1, 0, 2, 0, 5, 0, 14, 0, 42.
- `mixed_moments`: sample mixed moments of X_0, X_1 .
- `word1`: $\|X_0\|_2^2 = 1$ and $\|X_0\|_4^4 = 2$.
- `words_234`: $\|X_0 \cdots X_{d-1}\|_4^4 = d + 1$ and $\|\cdot\|_2^2 = 1$ for $d = 2, 3, 4$.
- `first_chaos_three`: for all integers a, b, c , $S = aX_0 + bX_1 + cX_2$ satisfies $\tau(S^2) = a^2 + b^2 + c^2$ and $\tau(S^4) = 2(a^2 + b^2 + c^2)^2$. This is a polynomial identity, so it holds for real coefficients too.
- `first_chaos_never_formula`: no nonzero such S has ratio $9/4$, $3/2$ or $81/16$.
- `Clause d` encodes $M_4(d)^2 = f(d)$ as $N(d)^2 \|W_d\|_2^4 = \|W_d\|_4^4 D(d)^2$, where $f = N/D$. Since $f(d) > 0$, $f(d) = \sqrt{y}$ is equivalent to $f(d)^2 = y$, so no real numbers are needed.
- `clause_false_d1`: $\neg \text{Clause } 1$.
- `clause_false_small`: `Clause` and the variants `ClauseM4`, `ClauseM4pow4` fail for $d = 1, \dots, 4$.
- `conjecture_00000007739_false (C2)` : $\neg(\text{Conj} \wedge C_2)$, where $\text{Conj} := \forall d \geq 1, \text{Clause } d$.

There is no `sorry` and no `native_decide`. The axioms used are at most `propext` and `Quot.sound`.

Limitations. Lean works on the finitely supported vectors of the Fock space; all vectors reached from Ω are finitely supported. The following are cited or proved in this report only:

- the passage to the von Neumann algebra and L^p norms (via (1));
- freeness of $X_0, X_1, \dots$ (Voiculescu's theorem);
- complex coefficients;
- general d (Proposition 3, Corollary 4);
- the higher-chaos and unitary readings.

Python (verify.py, standard library). Exact rational and Gaussian-rational arithmetic. Moments are computed by two independent methods: sparse Fock-space operators, and counting monochromatic non-crossing pairings. The two agree on all 9841 words of length ≤ 8 in three letters. The script checks:

- $\|W_d\|_4^4 = d + 1$ for $d = 1..5$ by both methods and by the Fuss–Catalan formula;
- all three normalisations for $d = 1..5$;
- the 2-adic argument for $d < 200$;
- ratio 2 on random real and complex first-chaos elements, including circular and elliptic ones;
- ratio ≥ 2 on random second-chaos elements, and ratio $2 + (d - 1)/n$ for sums of free copies;
- the Haar-unitary formula.